

Product Backlog Validation

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Document Purpose

This document validates the Product Backlog for the American Latin Class Attendance System. It connects each backlog item with its related user stories, functional requirements, acceptance criteria, implemented endpoints and verification evidence.

The objective is to demonstrate that the backlog was not only listed as planning work, but also implemented and validated through tests, Postman evidence, API documentation, database records and AWS deployment.

Validation Scope

Area	Evidence
Requirements source	<code>02Requirements/requirements.md</code>
API validation report	<code>03Documentation/api-validation-report.md</code>
Business rules report	<code>03Documentation/business-rules-api-test-report.md</code>
Postman collection	<code>postman/American-Latin-Class-API.postman_collection.json</code>
Python analytics collection	<code>postman/American-Latin-Class-Analytics-API.postman_collection.json</code>
JWT/Postman evidence	<code>postman/evidence/postman-local-jwt-auth-evidence.md</code>
Python API evidence	<code>postman/evidence/python-analytics-api-evidence.md</code>
Cache management evidence	<code>03Documentation/cache-management.md</code> , <code>03Documentation/api-validation-report.md</code>
Database evidence	<code>06Code/migrations/001_initial_schema.sql</code> , <code>06Code/seeder/001_seed.sql</code>
Deployment evidence	<code>03Documentation/aws-deployment-guide.md</code> , <code>07Other/nginx-alc-frontend.conf</code>

Backlog Validation Summary

Metric	Result
Original prioritized backlog items	10

Metric	Result
Additional validated extension items	2
Validated backlog items	12
Items with endpoint/API evidence	12
Items with automated or manual test evidence	12
Final validation status	Completed

Backlog Item Validations

PBI-01 - Auth, Sessions, RBAC And Private Pages

Field	Validation
Related stories / requirements	US-02, US-07, FR-02, FR-03, FR-04
Acceptance criteria	Existing academy users can open a secure app session with email/password or linked Google Sign-In; JWT Bearer access works in Postman; logout revokes the session; protected routes reject anonymous or revoked tokens.
Endpoint evidence	<code>POST /api/v1/auth/login</code> , <code>POST /api/v1/auth/google</code> , <code>GET /api/v1/auth/me</code> , <code>POST /api/v1/auth/logout</code>
Documentation evidence	<code>03Documentation/oauth-session.md</code> , <code>03Documentation/postman-token-auth-proof.md</code>
Status	Completed / Validated

PBI-02 - PostgreSQL Schema, Seeders And Repository Abstraction

Field	Validation
Related requirements	FR-05 through FR-14, NFR-03
Acceptance criteria	Normalized PostgreSQL schema exists; seed data loads roles, permissions, branches, dance categories/styles and rules; repository layer isolates persistence.
Code evidence	<code>06Code/migrations/001_initial_schema.sql</code> , <code>06Code/seeder s/001_seed.sql</code> , <code>06Code/prisma/schema.prisma</code> , <code>06Code/src/repositories</code>
Status	Completed / Validated

PBI-03 - Academic Catalog Management APIs

Field	Validation
Related stories / requirements	US-04, FR-05, FR-06
Acceptance criteria	Authorized users can list academic catalog resources; directors/admin can create and update academic data.
Endpoint evidence	<code>/api/v1/branches</code> , <code>/api/v1/students</code> , <code>/api/v1/teachers</code> , <code>/api/v1/dance-categories</code> , <code>/api/v1/dance-styles</code> , <code>/api/v1/class-groups</code> , <code>/api/v1/class-sessions</code>
Test evidence	<code>03Documentation/api-validation-report.md</code>
Status	Completed / Validated

PBI-04 - Attendance And Teacher Check-In Workflows

Field	Validation
Related stories / requirements	US-03, FR-07, FR-08
Acceptance criteria	Attendance can be recorded once per student/session; duplicate attendance is rejected; teacher check-in opens one active record and check-out closes it.
Endpoint evidence	<code>POST /api/v1/student-attendance</code> , <code>POST /api/v1/teacher-attendance/check-in</code> , <code>PATCH /api/v1/teacher-attendance/{id}/check-out</code>
Test evidence	<code>03Documentation/business-rules-api-test-report.md</code>
Status	Completed / Validated

PBI-05 - Absence Justifications And Audit Logs

Field	Validation
Related stories / requirements	US-04, US-06, FR-09, FR-14
Acceptance criteria	Authorized users submit absence justifications; directors review them; important academic actions are audited.
Endpoint evidence	<code>/api/v1/absence-justifications</code> , <code>/api/v1/absence-justifications/{id}/review</code> , <code>/api/v1/audit-logs</code>
Test evidence	<code>03Documentation/api-validation-report.md</code>

Field	Validation
Status	Completed / Validated

PBI-06 - Scholarship And Level Promotion Workflows

Field	Validation
Related stories / requirements	US-05, FR-10, FR-11
Acceptance criteria	Scholarship candidates require at least 90 percent attendance; promotion applies to B1 to B2; directors register evaluation evidence.
Endpoint evidence	<code>/api/v1/reports/scholarships/{studentId}/candidate</code> , <code>/api/v1/scholarship-evaluations</code> , <code>/api/v1/reports/level-promotions/{studentId}/candidate</code> , <code>/api/v1/level-promotion-evaluations</code>
Test evidence	<code>03Documentation/business-rules-api-test-report.md</code>
Status	Completed / Validated

PBI-07 - Reports By Branch, Teachers And Consolidated Roles

Field	Validation
Related stories / requirements	US-06, FR-12, FR-13
Acceptance criteria	GeneralDirector/Admin can access branch summaries, teacher payment reports and restricted reports.
Endpoint evidence	<code>/api/v1/reports/branches/summary</code> , <code>/api/v1/reports/teachers/{teacherId}/payment</code>
Documentation evidence	<code>03Documentation/api-documentation.md</code> , <code>03Documentation/api-validation-report.md</code>
Status	Completed / Validated

PBI-08 - Landing Page, Enrollment Form And Private Dashboard

Field	Validation
Related stories / requirements	US-01, US-02, FR-01

Field	Validation
Acceptance criteria	Visitor can view academy information and submit enrollment; authenticated users can enter the private dashboard; private pages redirect after logout/no session.
Code evidence	React + Vite frontend under <code>06Code/frontend</code>
Endpoint evidence	<code>POST /api/v1/enrollment-requests</code>
Status	Completed / Validated

PBI-09 - AWS Deployment Documentation And Diagrams

Field	Validation
Related requirement	NFR-10
Acceptance criteria	AWS architecture uses EC2 instances, RDS PostgreSQL and Nginx/HTTPS routing; documentation explains deployment and network responsibilities.
Documentation evidence	<code>03Documentation/aws-deployment-guide.md</code> , <code>04UMLDiagrams/aws-deployment.puml</code> , <code>070ther/nginx-alc-frontend.conf</code>
Runtime evidence	Deployed EC2/RDS services
Status	Completed / Validated

PBI-10 - Final Test Evidence And Requirement Reevaluation

Field	Validation
Related requirements	All requirements
Acceptance criteria	Automated and manual evidence exists for auth, CRUD, attendance, reports, sessions, business rules and deployed APIs.
Evidence	<code>03Documentation/api-validation-report.md</code> , <code>03Documentation/business-rules-api-test-report.md</code> , <code>postman/evidence/</code> , <code>08PDFDeliverables</code>
Status	Completed / Validated

PBI-11 - Python FastAPI Analytics API Extension

Field	Validation
Related stories / requirements	Additional class activity, FR-07, FR-10, FR-12, FR-13
Acceptance criteria	A Python API exists for the project; it is deployed separately, protected by the same JWT session, and calculates real academic analytics.
Code evidence	<code>06Code/python-analytics-api</code>
Endpoint evidence	<code>/api/analytics/v1/health</code> , <code>/students/{student_id}/attendance-risk</code> , <code>/students/{student_id}/scholarship-readiness</code> , <code>/branches/{branch_id}/performance-summary</code> , <code>/teachers/{teacher_id}/workload-summary</code>
Test evidence	<code>postman/evidence/python-analytics-api-evidence.md</code>
Status	Completed / Validated

PBI-12 - Controlled Cache Management Extension

Field	Validation
Related stories / requirements	US-08, FR-15, NFR-12, BR-14
Acceptance criteria	Public, private and no-store cache policies are explicit; repeated safe reads expose memory-cache <code>MISS/HIT</code> ; state-changing academic actions invalidate affected cache tags; scoped reports are cached per authenticated actor.
Code evidence	<code>06Code/src/services/CacheService.js</code> , <code>06Code/src/middleware/cacheControl.js</code> , <code>06Code/src/routes/api.js</code> , <code>070the r/nginx-alc-frontend.conf</code>
Endpoint evidence	<code>/api/v1/auth/config</code> , <code>/api/v1/roles</code> , <code>/api/v1/branches</code> , <code>/api/v1/reports/branches/summary</code> , <code>/api/analytics/v1/health</code>
Test evidence	<code>06Code/tests/integration/cache.test.js</code> , <code>03Documentation/api-validation-report.md</code>
Status	Completed / Validated

Requirement Coverage

Requirement	Covered By Backlog Items	Validation Result
FR-01 Public landing and enrollment request	PBI-08	Validated
FR-02 Google OAuth registration/login	PBI-01	Validated
FR-03 Secure session management	PBI-01	Validated
FR-04 Role-based authorization	PBI-01	Validated
FR-05 Core academic management	PBI-03	Validated
FR-06 Dance category/style management	PBI-03	Validated
FR-07 Student attendance	PBI-04, PBI-11	Validated
FR-08 Teacher check-in/check-out	PBI-04	Validated
FR-09 Absence justifications	PBI-05	Validated
FR-10 Scholarship eligibility/evaluation	PBI-06, PBI-11	Validated
FR-11 Level promotion eligibility/evaluation	PBI-06	Validated
FR-12 Teacher hours/payment report	PBI-07, PBI-11	Validated
FR-13 Branch/consolidated reports	PBI-07, PBI-11	Validated
FR-14 Audit logs	PBI-05	Validated
FR-15 Controlled cache management	PBI-12	Validated
NFR-12 Controlled HTTP and in-memory cache behavior	PBI-12	Validated

Validation Methods

Method	Command / Tool	Result
Node.js automated tests	<code>cd 06Code && npm test</code>	5 suites passed, 24 tests passed
API validation suite	<code>cd 06Code && npm run test:api:validation</code>	114/114 validation cases passed
Python API tests	<code>cd 06Code/python-analytics-api && .\.venv\Scripts\python.exe -m unittest discover -s tests -v</code>	9 tests passed

Method	Command / Tool	Result
Manual Postman verification	Main API collection and Python analytics collection	Login, token use, API calculations and logout validated
AWS deployment verification	HTTPS routes through Nginx and EC2 services	Node API and Python API reachable through public domain
PDF deliverables	<code>node 070ther/pdf-tools/generate-pdfs.js</code>	Documentation package generated

Definition Of Done Validation

Definition Of Done Item	Evidence	Status
Requirement is documented	<code>02Requirements/requirements.md</code>	Done
API endpoint or UI workflow exists	<code>03Documentation/api-documentation.md</code> , <code>06Code/src</code> , <code>06Code/frontend</code> , <code>06Code/python-analytics-api</code>	Done
Business rule is implemented in service layer	<code>06Code/src/services</code> , <code>06Code/python-analytics-api/app/services.py</code>	Done
Data model supports the feature	PostgreSQL migration, Prisma schema, RDS seed records	Done
Automated or manual test evidence exists	Jest, API validation suite, Python unittest, Postman evidence	Done
Deployment path is documented	AWS guide and Nginx configuration	Done
Final documentation exists as Markdown and PDF	<code>03Documentation</code> , <code>postman/evidence</code> , <code>08PDFDeliverables</code>	Done

Final Conclusion

The Product Backlog is fully validated for the current academic submission. Each backlog item is connected to implemented code, REST endpoints, database support, tests, Postman evidence or AWS deployment documentation.

The additional Python FastAPI analytics API is also validated as an extension item. It demonstrates a separate Python backend service that reads real project data, calculates academic indicators and reuses the existing JWT session model.

The controlled cache management extension is validated through headers, memory-cache hit/miss evidence, invalidation checks and deployment-ready Nginx cache rules.